

Experiment 4: Total Sales and Percentage Increase

AIM

To calculate the total sales for four quarters and determine the percentage increase in sales from the first quarter to the fourth quarter using NumPy.

ALGORITHM

1. Import NumPy.
2. Create a NumPy array containing sales for four quarters.
3. Calculate total annual sales using `np.sum()`.
4. Store the first-quarter sales and fourth-quarter sales.
5. Calculate the percentage increase using the formula:
6. Display the total sales and percentage increase.

CODE

```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)

first_quarter = sales_data[0]

fourth_quarter = sales_data[3]

percentage_increase = (

    (fourth_quarter - first_quarter) / first_quarter

) * 100

print("Total sales for the year:", total_sales)

print("Percentage increase from Q1 to Q4:",

    percentage_increase, "%")
```

INPUT

Q1 = 10000

Q2 = 12000

Q3 = 15000

Q4 = 18000

OUTPUT

```
import numpy as np
sales_data = np.array([10000, 12000, 15000, 18000])
total_sales = np.sum(sales_data)
first_quarter = sales_data[0]
fourth_quarter = sales_data[3]
percentage_increase = (
    (fourth_quarter - first_quarter) / first_quarter
) * 100
print("Total sales for the year:", total_sales)
print("Percentage increase from Q1 to Q4:",
      percentage_increase, "%")
```

```
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:1
3:20) [MSC v.1940 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for
more information.
>>>
= RESTART: C:/Users/monik/Desktop/SLOT B & C/FODS/expl
.py
Total sales for the year: 55000
Percentage increase from Q1 to Q4: 80.0 %
>>>
```

RESULT

Thus, the total sales for the year were calculated as 55000, and the percentage increase from the first quarter to the fourth quarter was 80%.